

TORINO CASELLE, Italy

WMO: 160590

Lat: 45.201N

Lon: 7.650E

Elev: 301

StdP: 97.76

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.0	-3.8	-13.0	1.3	1.3	-10.9	1.5	0.8	8.7	11.1	6.5	8.5	1.4	300	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	31.2	22.4	30.0	21.7	28.9	21.1	23.8	29.1	22.9	28.2	22.1	27.0	2.3	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.2	17.5	26.2	21.2	16.4	25.6	20.2	15.5	25.1	72.7	29.2	69.3	28.4	66.3	26.9	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.1	4.2	-7.5	33.3	2.3	1.6	-9.1	34.5	-10.5	35.4	-11.7	36.4	-13.4	37.5		
(5)			-8.5	24.8	2.0	1.1	-10.0	25.7	-11.2	26.3	-12.3	27.0	-13.8	27.8	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	2.9	4.6	9.0	12.5	16.8	21.1	23.2	22.6	18.4	13.6	7.7	3.5
	DBStd	7.73	2.76	3.21	3.36	3.19	2.99	3.05	2.41	2.54	2.80	3.12	3.03	2.91
	HDD10.0	728	220	152	57	12	0	0	0	0	0	8	79	201
	HDD18.3	2360	478	384	288	176	64	9	1	2	32	149	318	459
	CDD10.0	1843	0	1	28	87	212	332	409	391	253	118	11	0
	CDD18.3	434	0	0	0	1	18	91	152	134	35	2	0	0
	CDH23.3	3302	0	0	2	14	135	683	1244	1014	200	9	0	0
	CDH26.7	891	0	0	0	1	19	181	366	296	28	1	0	0
Wind	WSAvg	1.8	1.4	1.6	2.0	2.1	2.0	2.0	2.1	2.0	1.7	1.4	1.5	1.4
Precipitation	PrecAvg	894	38	53	77	89	118	103	63	73	75	91	67	49
	PrecMax	1617	153	245	186	257	278	176	252	233	228	285	229	202
	PrecMin	279	0	0	1	1	8	39	0	2	2	0	3	0
	PrecStd	287	39	62	53	64	85	44	56	53	66	81	60	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	18.0	22.9	25.1	29.0	32.2	32.8	33.2	29.1	24.2	18.0	14.0
		MCWB	8.0	10.7	12.7	15.9	19.1	22.6	23.1	23.6	20.1	16.6	11.5	7.6
	2%	DB	11.9	14.2	19.8	22.6	26.5	30.2	31.1	31.1	27.2	22.0	15.8	11.2
		MCWB	6.5	8.2	11.5	14.5	18.3	22.1	22.5	22.5	19.4	15.9	10.9	6.9
	5%	DB	9.5	12.2	17.8	20.9	24.8	28.9	30.0	29.8	25.8	20.2	13.9	9.8
		MCWB	5.2	6.7	10.8	13.6	17.3	21.3	21.7	21.7	18.4	15.0	10.3	6.4
	10%	DB	7.9	10.2	15.8	19.0	23.0	27.1	28.9	28.1	24.0	18.8	12.3	8.1
		MCWB	4.5	5.8	9.7	12.6	16.3	20.3	21.2	20.6	17.6	14.4	9.7	5.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	10.7	13.5	16.8	20.6	24.9	25.0	24.7	21.9	18.1	13.5	9.8
		MCDB	12.0	16.2	19.8	23.3	26.8	29.8	29.5	30.9	25.8	21.8	16.3	11.1
	2%	WB	7.1	8.8	12.3	15.3	18.9	23.2	23.8	23.5	20.7	17.1	12.2	8.1
		MCDB	9.7	12.9	17.9	21.0	25.0	28.2	28.8	29.2	25.1	20.2	14.2	10.0
	5%	WB	5.8	7.6	11.2	14.2	18.0	22.2	22.9	22.7	19.7	16.3	11.4	6.8
		MCDB	8.2	10.6	16.3	19.6	23.5	27.0	28.3	27.9	24.0	19.0	13.1	8.5
	10%	WB	4.7	6.4	10.2	13.1	17.0	21.2	22.2	21.8	18.7	15.4	10.4	5.9
		MCDB	6.8	9.0	14.4	17.9	21.8	26.0	27.1	26.4	22.8	17.9	12.0	7.5
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.2	10.2	9.5	9.3	9.5	10.0	9.9	9.4	8.3	7.7	8.1
		MCDBR	11.9	13.0	14.0	12.8	11.9	11.0	11.1	11.5	11.5	10.7	10.3	10.8
		MCWBR	7.7	8.0	7.8	6.9	6.0	5.3	5.4	5.2	5.1	5.8	6.7	7.2
	5% WB	MCDBR	10.0	11.2	12.1	11.0	10.5	10.0	9.8	10.3	9.6	8.6	7.5	8.3
		MCWBR	7.1	7.4	7.2	6.2	5.7	5.1	5.2	5.2	4.7	5.1	5.3	6.1
Clear-Sky Solar Irradiance	taub	0.309	0.337	0.385	0.436	0.443	0.457	0.442	0.435	0.422	0.397	0.345	0.303	
	taud	2.406	2.330	2.224	2.115	2.164	2.176	2.222	2.249	2.252	2.291	2.391	2.434	
	Ebn at Noon	799	837	838	823	830	818	825	816	791	754	743	768	
	Edh at Noon	73	95	122	148	145	145	136	128	118	99	75	65	
All-Sky Solar Radiation	RadAvg	1.64	2.47	3.60	4.44	5.21	6.13	6.32	5.48	4.20	2.67	1.62	1.38	
	RadStd	0.19	0.31	0.45	0.51	0.45	0.45	0.43	0.33	0.32	0.33	0.24	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	+0.53	N/A	N/A	N/A	N/A	N/A	N/A	+105	+67	(46)
(47)	Regional (63 neighbors)	+0.30	N/A	N/A	+0.64	N/A	+0.24	N/A	+60	+18	(47)

Nomenclature: See separate page